Progetto controlli

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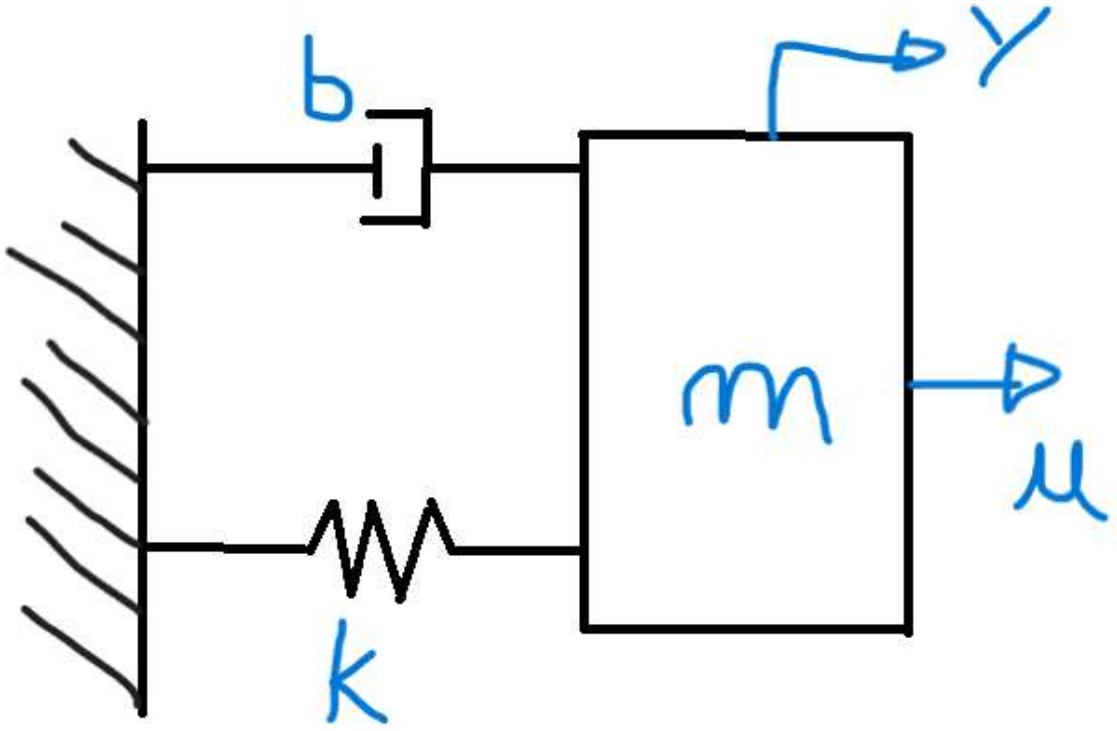
da control lare

- Variabile di controllo = u,forza agente sulla massa
- Variabile controllata = y, posizione della massa
- b =coefficiente di smorzamento
- m = massa
- k = costante



modelizzazione

- Tramite eq.di Eulero-Lagrange:

```
P =
                1
  ----------------
  s^2 + 1.5 s + 18
```

```
A =

         0    1.0000
  -18.0000   -1.5000


B =

     0
     1


C =

     1     0


D =

     0


P =

         1
  ----------------
  s^2 + 1.5 s + 18

Continuous-time transfer function.
```

Sistema caratteristiche

Figura 1

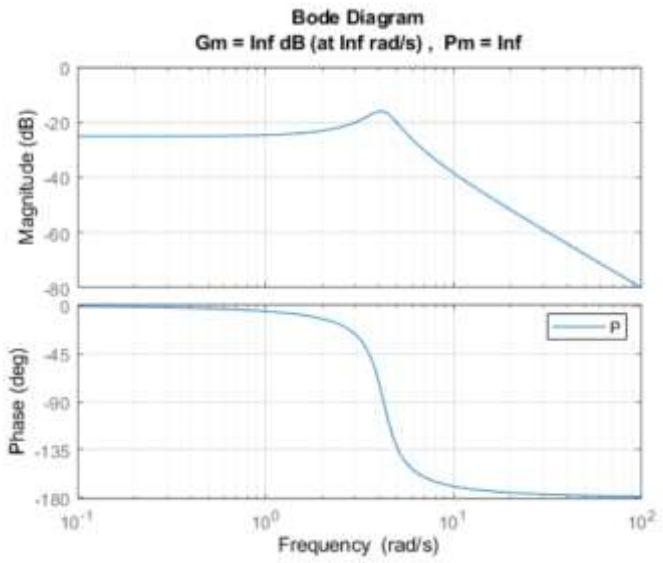


Figura 2

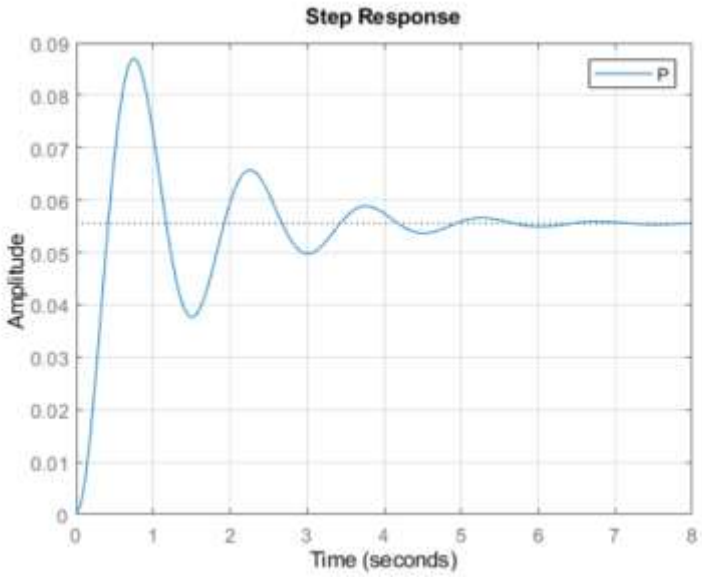
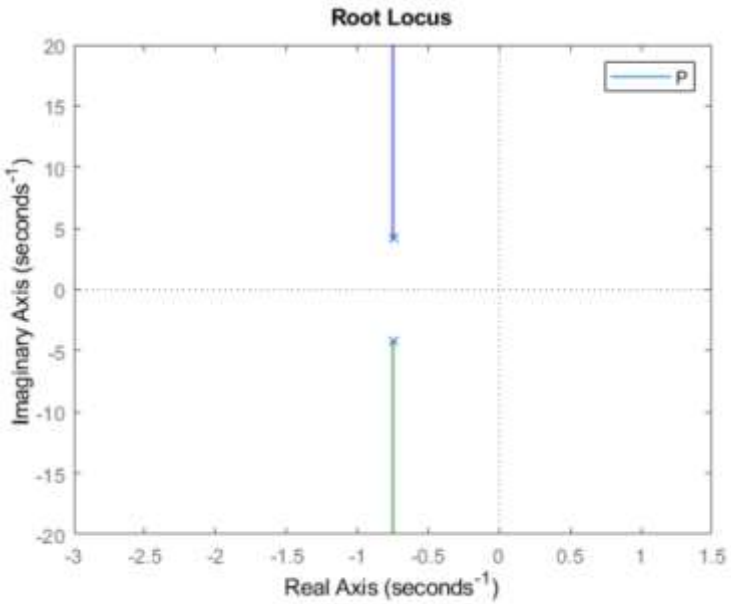


Figura 3



Zita $\simeq 0.17$ w_n $\simeq 4.24$

Richieste di progetto

- Errore nullo per riferimenti a rampa -> due poli nell’origine
- Attenuazione del 90% di un disturbo d1 agente sull’impianto ad una frequenza di 15Hz
- Tempo di assestamento inferiore a 3 secondi ->una w_t >6

Controllore in feedback proposto : prestazioni a regime

Due poli in zero+due zeri in -2

```
C1 =

  s^2 + 4 s + 4
  -------------
       s^2
```

Una rete anticipatrice con massimo anticipo di fase vicino a 6 rad/s

```
C2 =

  0.4392 s + 1
  -------------
  0.04392 s + 1
```

Effetti del controllore

Figura 4

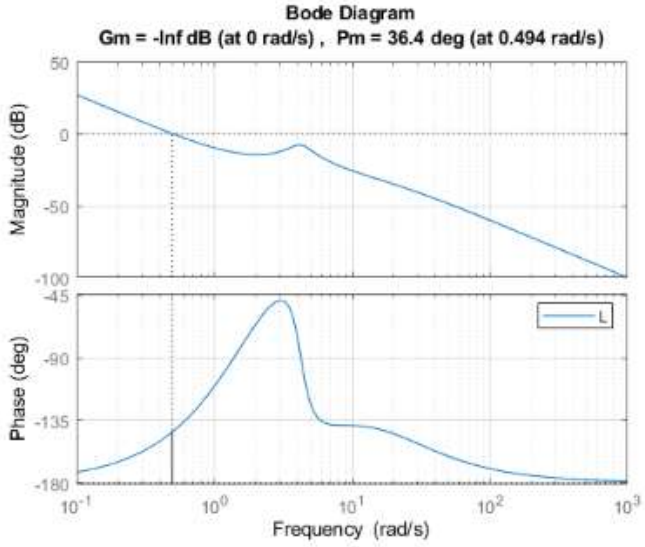


Figura 5

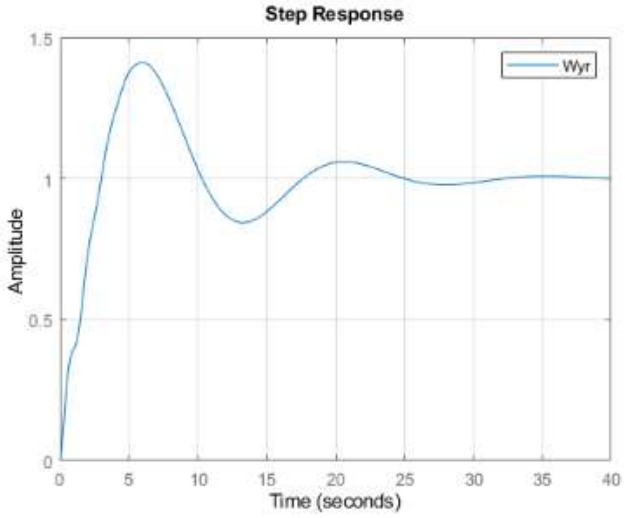
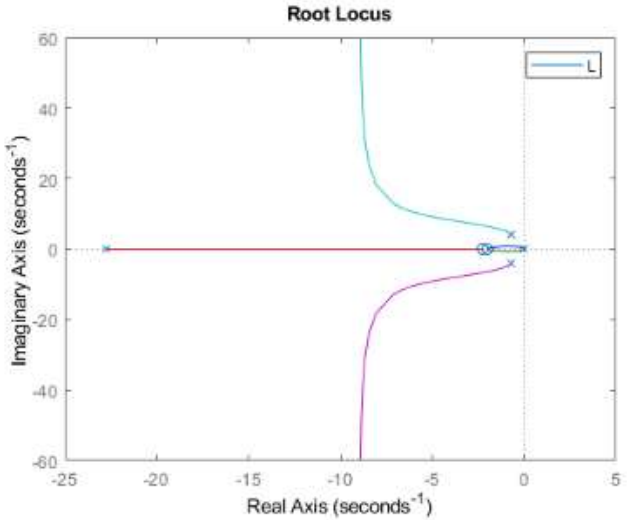


Figura 6



Controllore in feedback proposto: prestazioni nel transitorio

- C3 = K , con K = 20 ->aumento w t

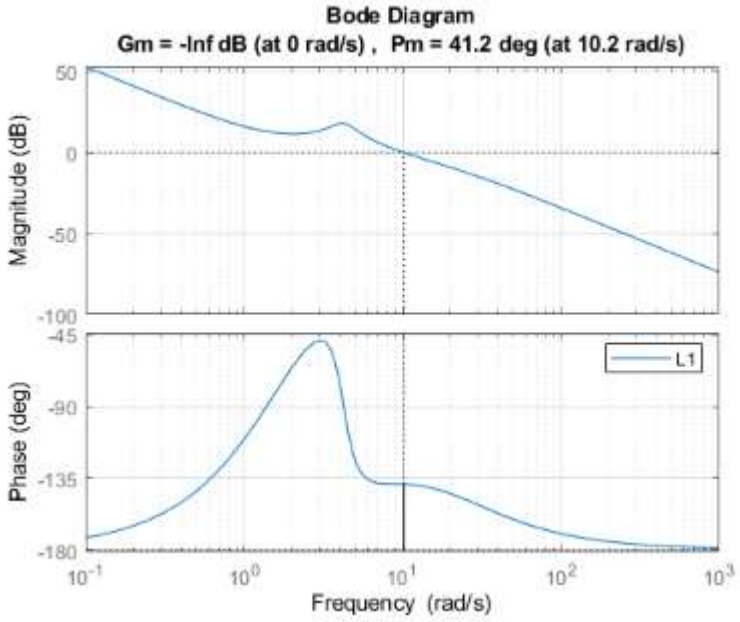


Figura 7

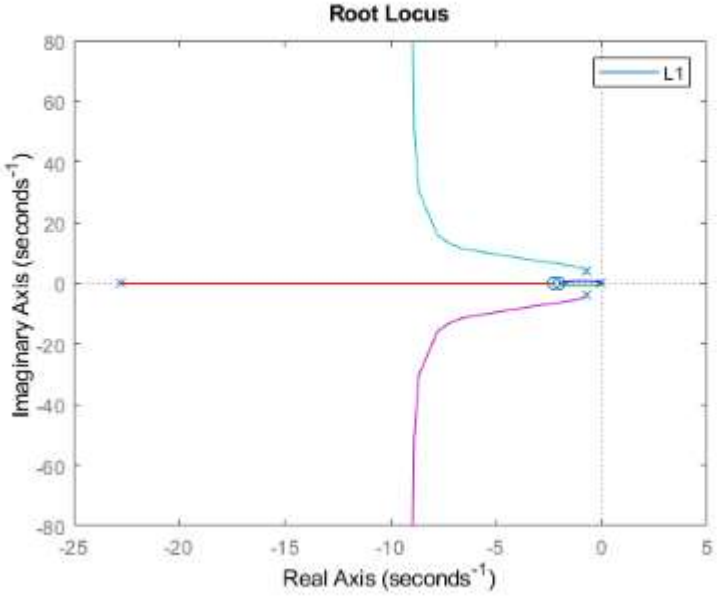


Figura 8

Aggiunta controllore in feedforward e analisi prestazioni

```
Ff =

  s^2 + 1.5 s + 18
 ------------------
 s^2 + 200 s + 1e04
```

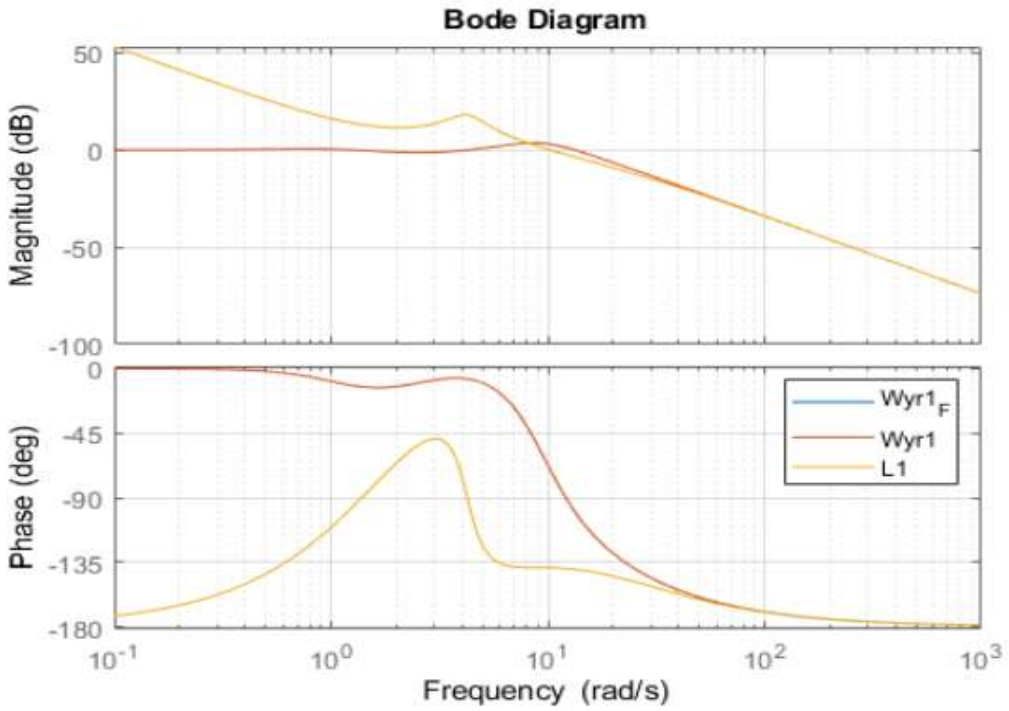


Figura 9

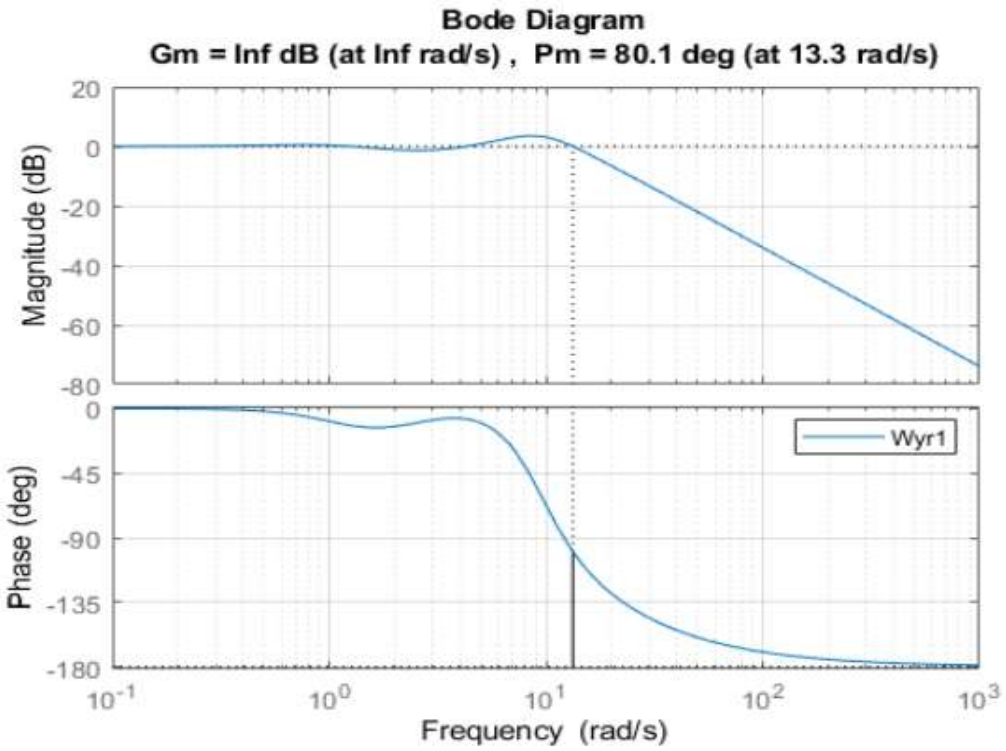
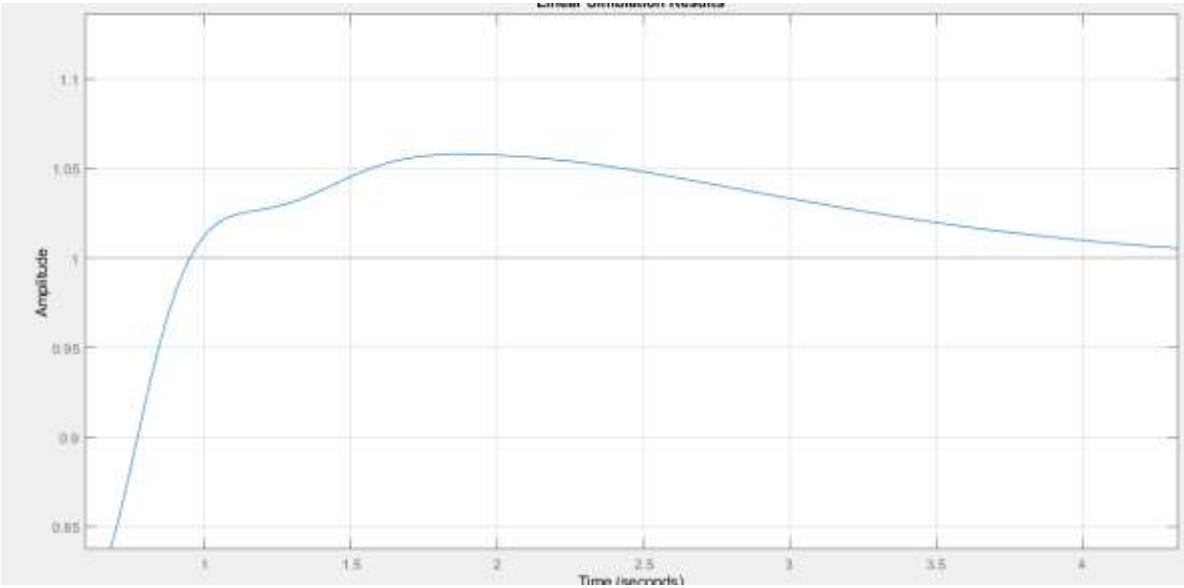
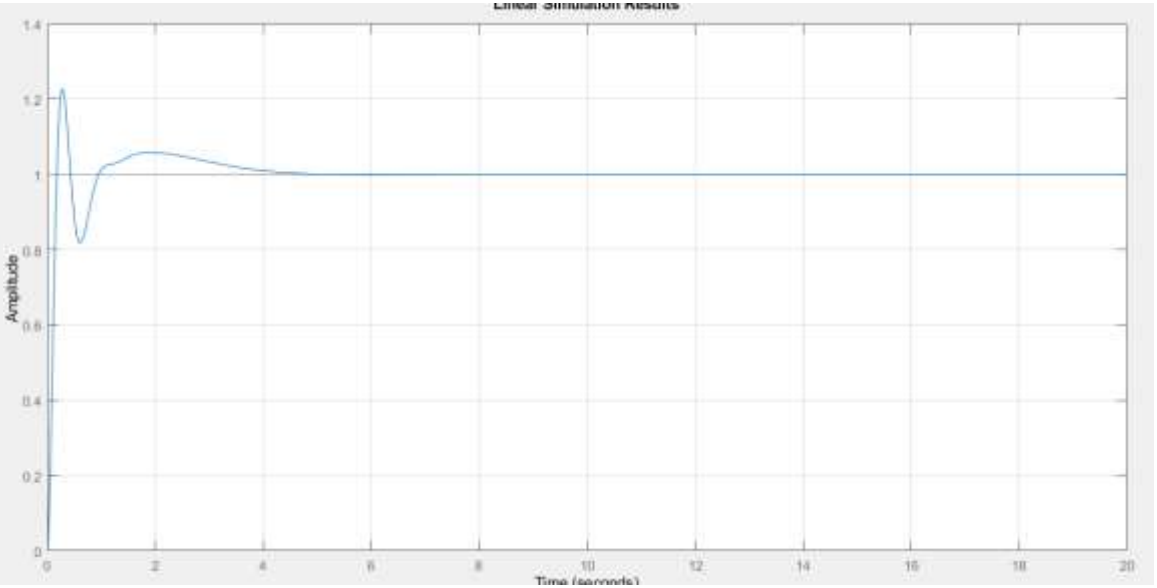
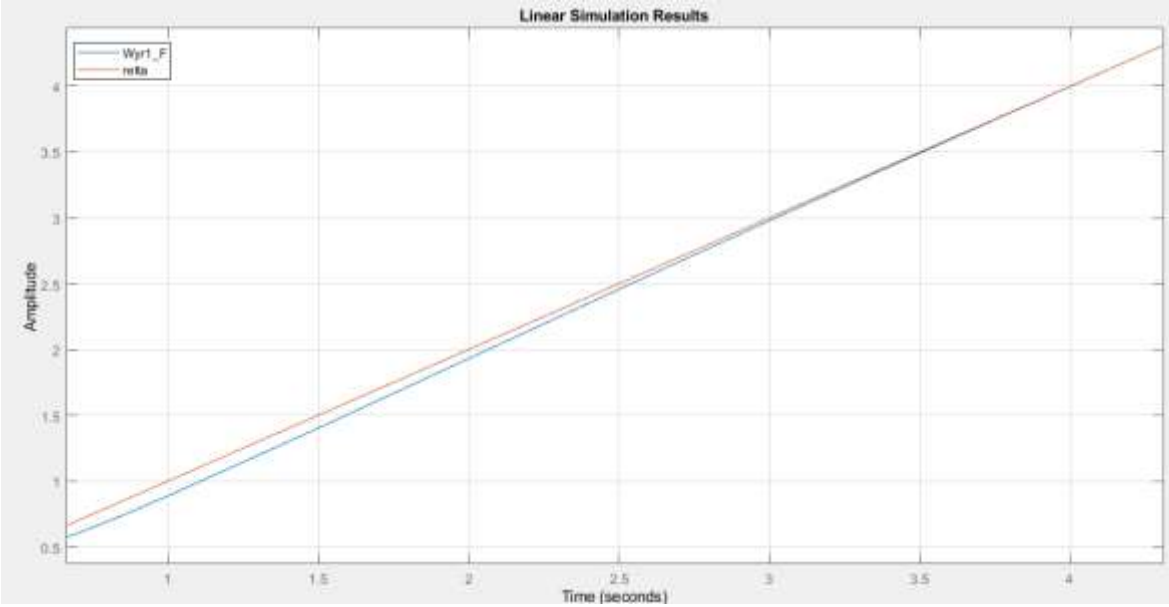


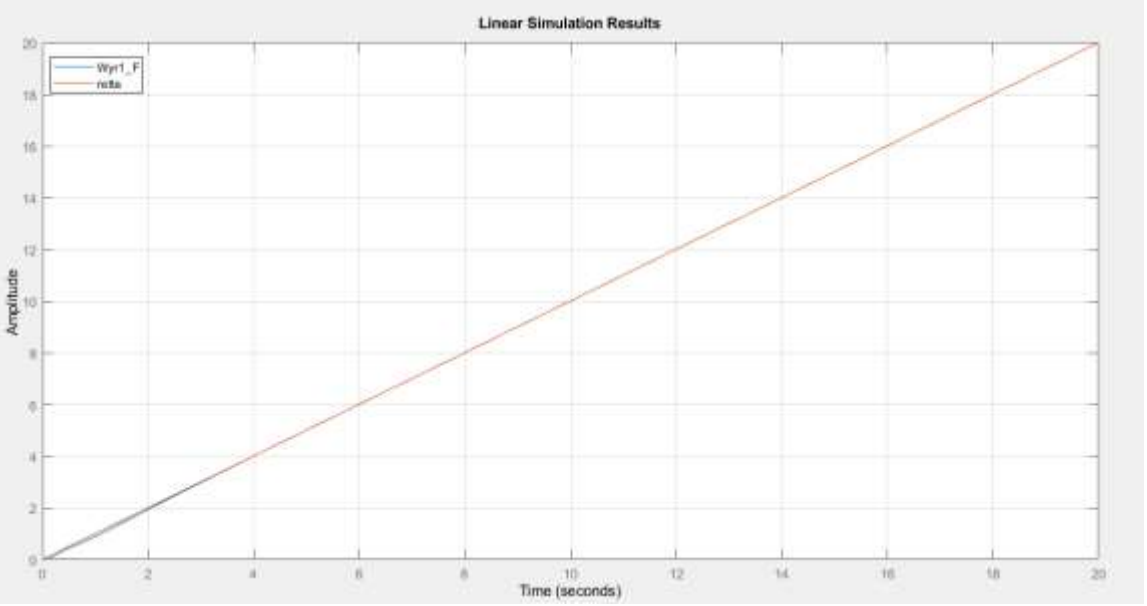
Figura 10

Simulazione a gradino e rampa,figure da11 a 14









Massimo ritardo ammissibile

- Pm $\simeq$ 41°, w_t $\simeq$ 10 rad/s -> T_max = 0.07
- Nel caso Pm = 180, T_max = 0.31

Analisi funzione di sensitività

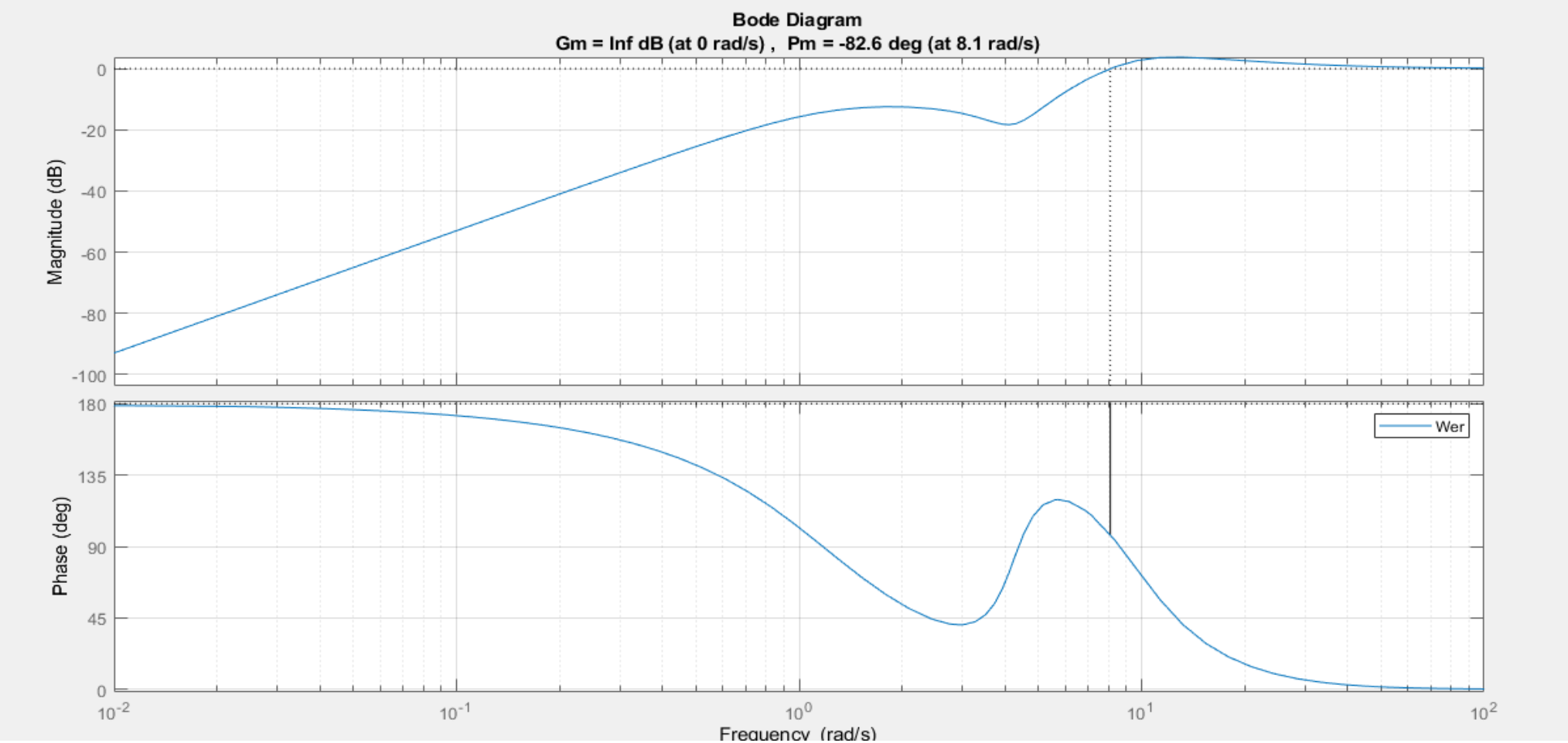


Figura 15

Analisi funzione di sensitività del controllo

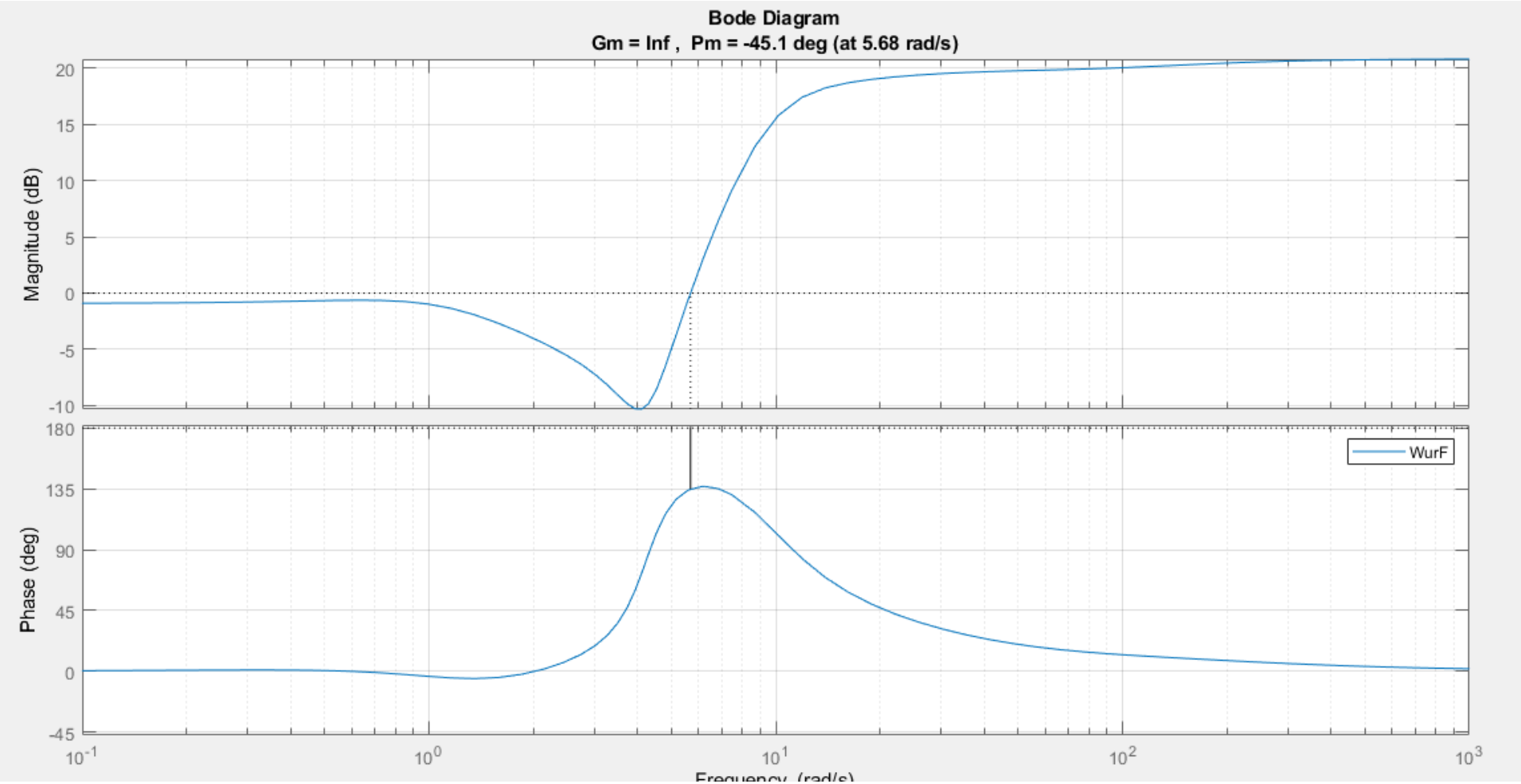


Figura 16

Variazioni dei parametri

- Ipotizzo la variazione dei parametri nei seguenti range:
- m ± 20% ->m ∈[0.8 ; 1.2]
- b ± 2% -> b ∈[16 ; 20]
- k ± 10% -> k ∈[1.35 ; 1.65]
- Applico Kharitonov e tabella di Routh.

Variazione dei parametri

Formula 1

```
s = tf('s')
m = 1.2;
k = 1.65;
b = 16;
poli_1 = (1/(s + 100)^2 + (20*(s + 2)^2*((5*2^(1/2)*5^(1/2)*s)/36 + 1))/(s^2*((2^(1/2)*5^(1/2)*s)/72 + 1)*(m*s^2 + k*s + b)))...
    /((20*(s + 2)^2*((5*2^(1/2)*5^(1/2)*s)/36 + 1))/(s^2*((2^(1/2)*5^(1/2)*s)/72 + 1)*(m*s^2 + k*s + b)) + 1);
p1 = minreal(poli_1);
[p1num,p1den] = tfdata(p1);
p1d = cell2mat(p1den);
myRouth(p1d)

m = 0.8;
k = 1.35;
b = 20;
poli_2 = (1/(s + 100)^2 + (20*(s + 2)^2*((5*2^(1/2)*5^(1/2)*s)/36 + 1))/(s^2*((2^(1/2)*5^(1/2)*s)/72 + 1)*(m*s^2 + k*s + b)))...
/((20*(s + 2)^2*((5*2^(1/2)*5^(1/2)*s)/36 + 1))/(s^2*((2^(1/2)*5^(1/2)*s)/72 + 1)*(m*s^2 + k*s + b)) + 1);
p2 = minreal(poli_2);
[p2num,p2den] = tfdata(p2);
p2d = cell2mat(p2den);
myRouth(p2d)

m = 1.2;
k = 1.35;
b = 20;
poli_3 = (1/(s + 100)^2 + (20*(s + 2)^2*((5*2^(1/2)*5^(1/2)*s)/36 + 1))/(s^2*((2^(1/2)*5^(1/2)*s)/72 + 1)*(m*s^2 + k*s + b)))...
    /((20*(s + 2)^2*((5*2^(1/2)*5^(1/2)*s)/36 + 1))/(s^2*((2^(1/2)*5^(1/2)*s)/72 + 1)*(m*s^2 + k*s + b)) + 1);
p3 = minreal(poli_3);
[p3num,p3den] = tfdata(p3);
p3d = cell2mat(p3den);
myRouth(p3d)


m = 0.8;
k = 1.65;
b = 16;
poli_4 = (1/(s + 100)^2 + (20*(s + 2)^2*((5*2^(1/2)*5^(1/2)*s)/36 + 1))/(s^2*((2^(1/2)*5^(1/2)*s)/72 + 1)*(m*s^2 + k*s + b)))...
    /((20*(s + 2)^2*((5*2^(1/2)*5^(1/2)*s)/36 + 1))/(s^2*((2^(1/2)*5^(1/2)*s)/72 + 1)*(m*s^2 + k*s + b)) + 1);
p4 = minreal(poli_4);
[p4num,p4den] = tfdata(p4);
p4d = cell2mat(p4den);
myRouth(p4d)
```

Variazione dei parametri

Web Browser - sistema da controllare

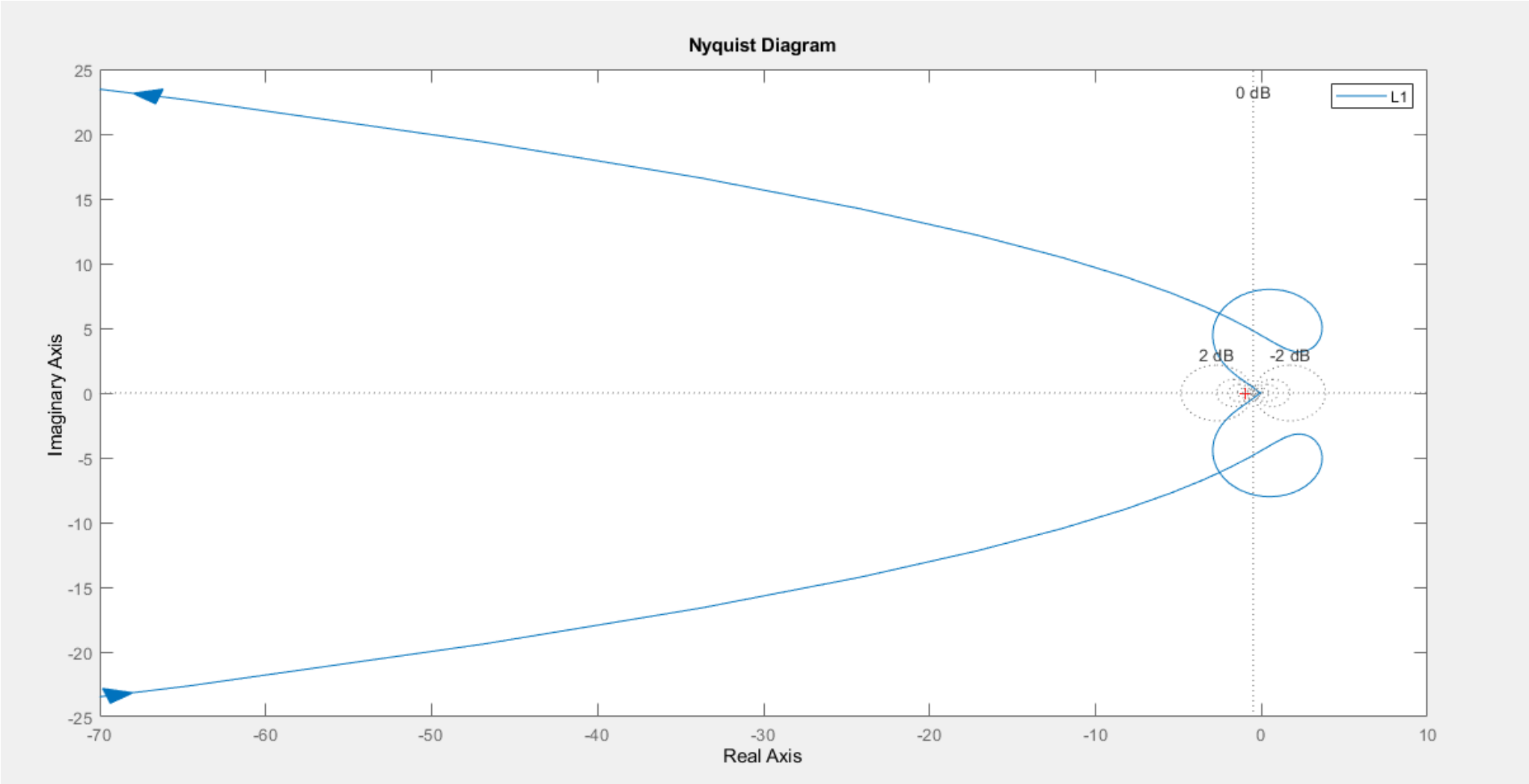
sistema da controllare

Location: file:///D:/Universit%C3%A0/controlli%20automatici/html/progettoControlli.html#11

```
[                                                
```

Stabilità assoluta

Figura 17



Schema finale e simulazione su simulink

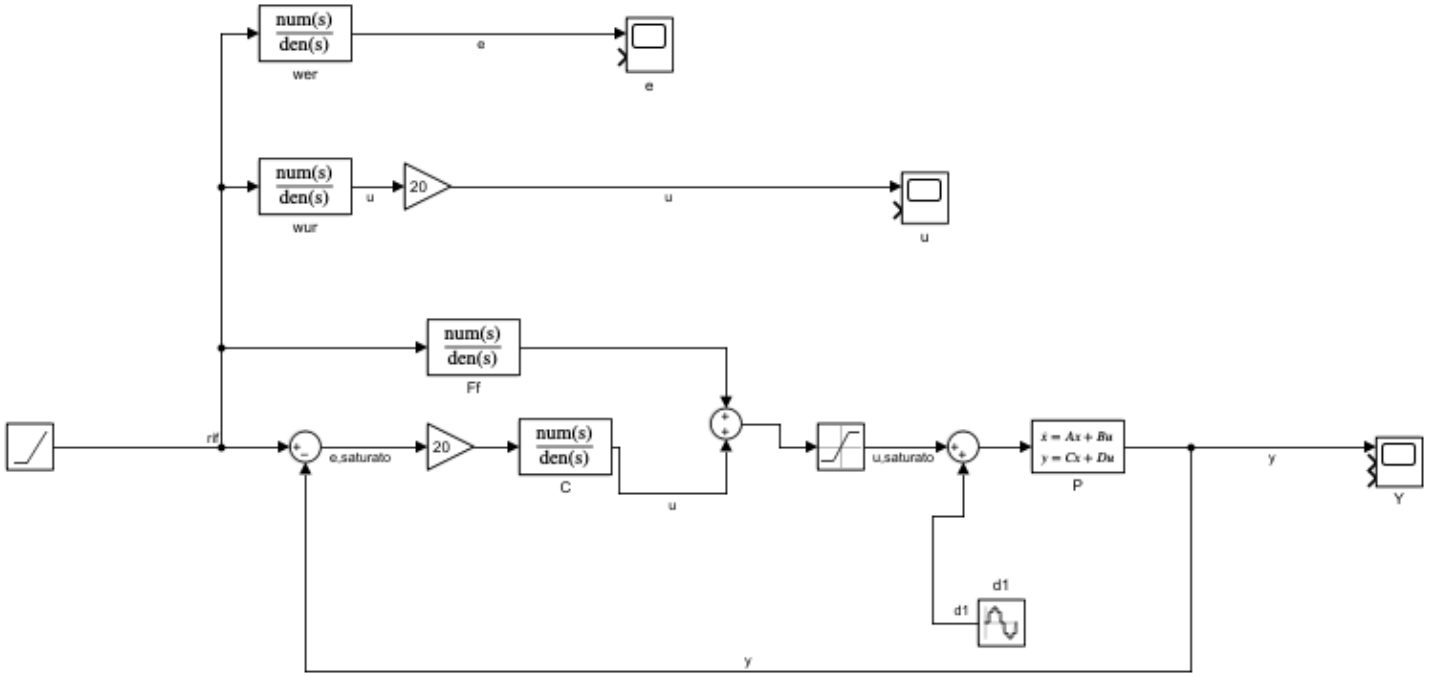


Figura 18

Uscita con ampiezza disturbo = 10, no saturazione

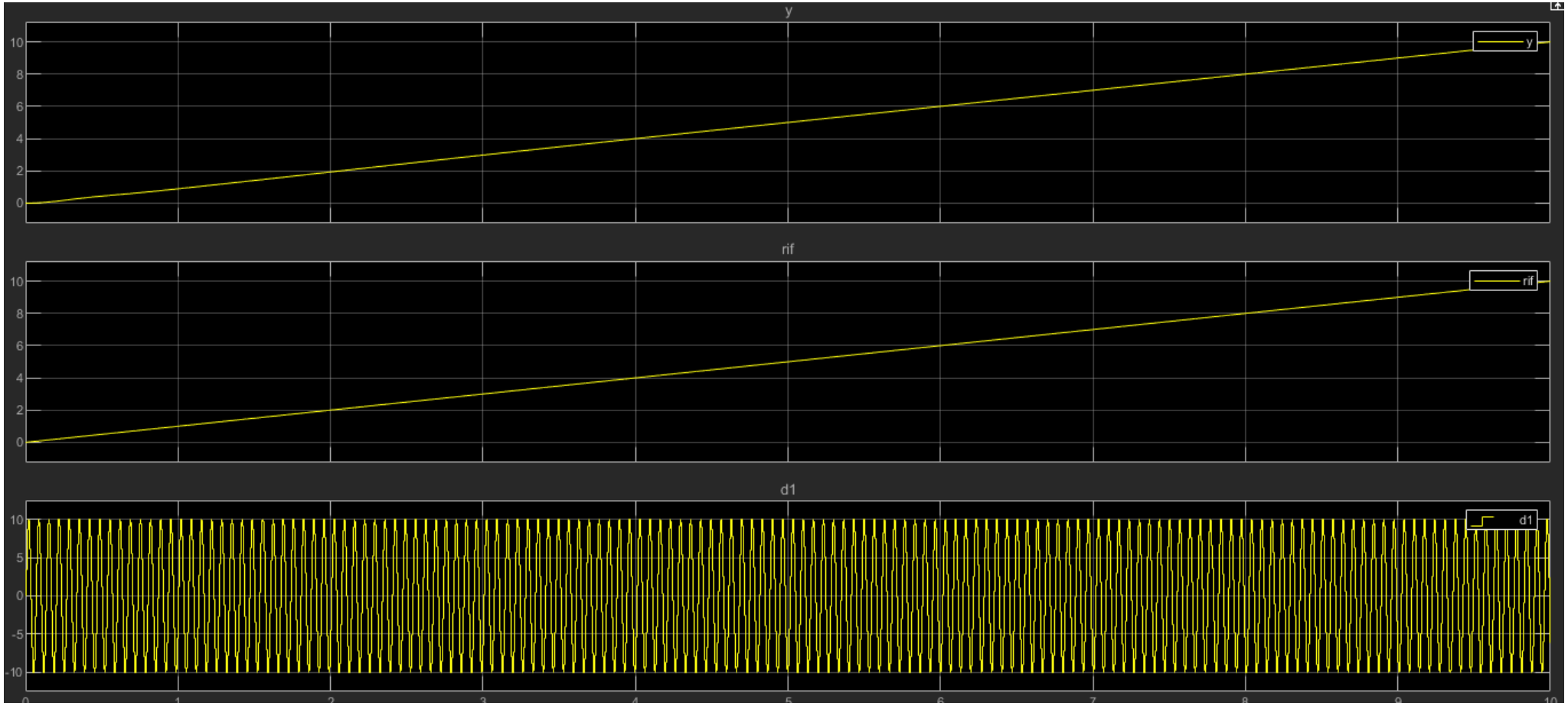


Figura 19

Controllo con ampiezza disturbo = 10, no saturazione

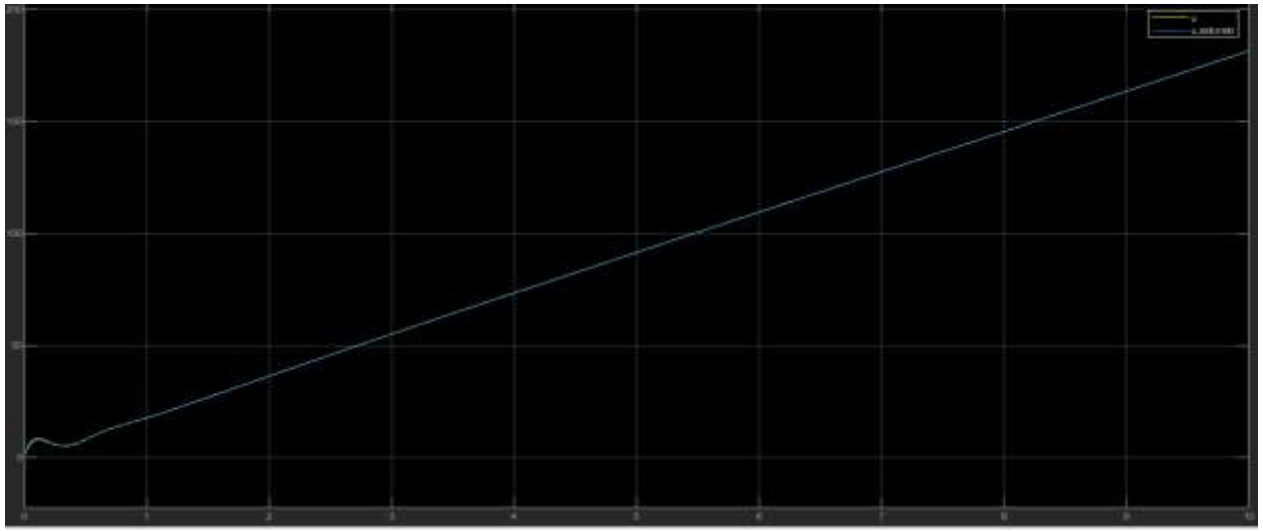
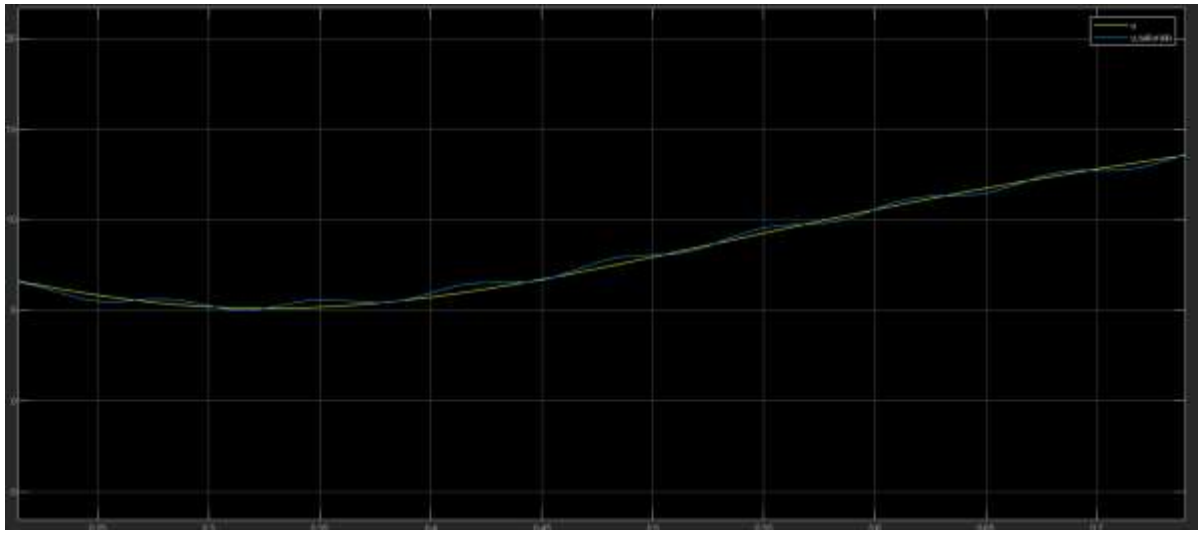


Figure 20 ↑ e 21→



Errore con ampiezza disturbo = 10, no saturazione

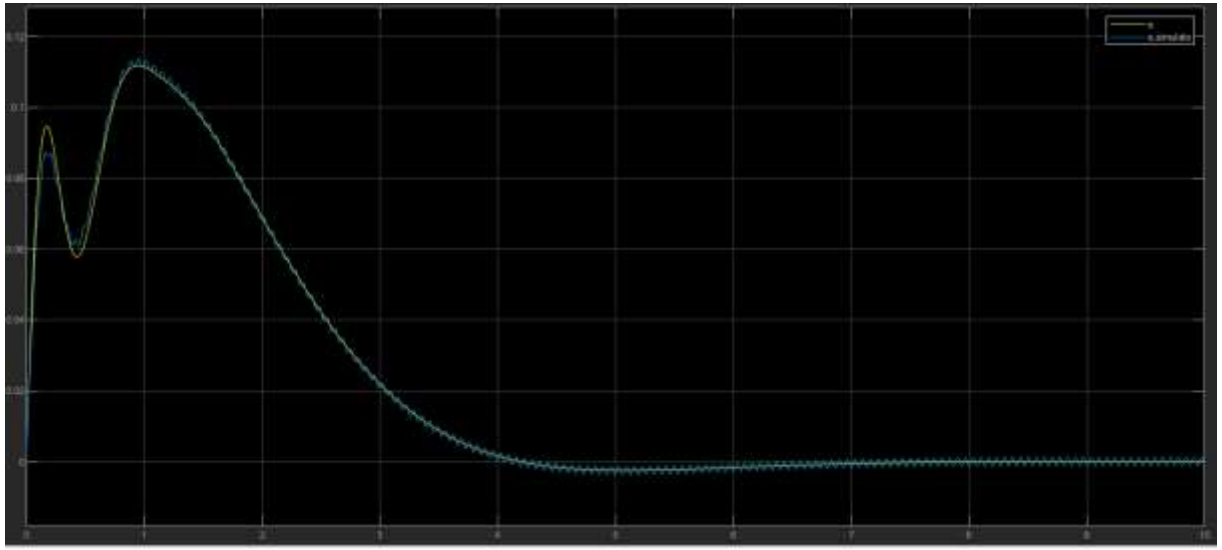
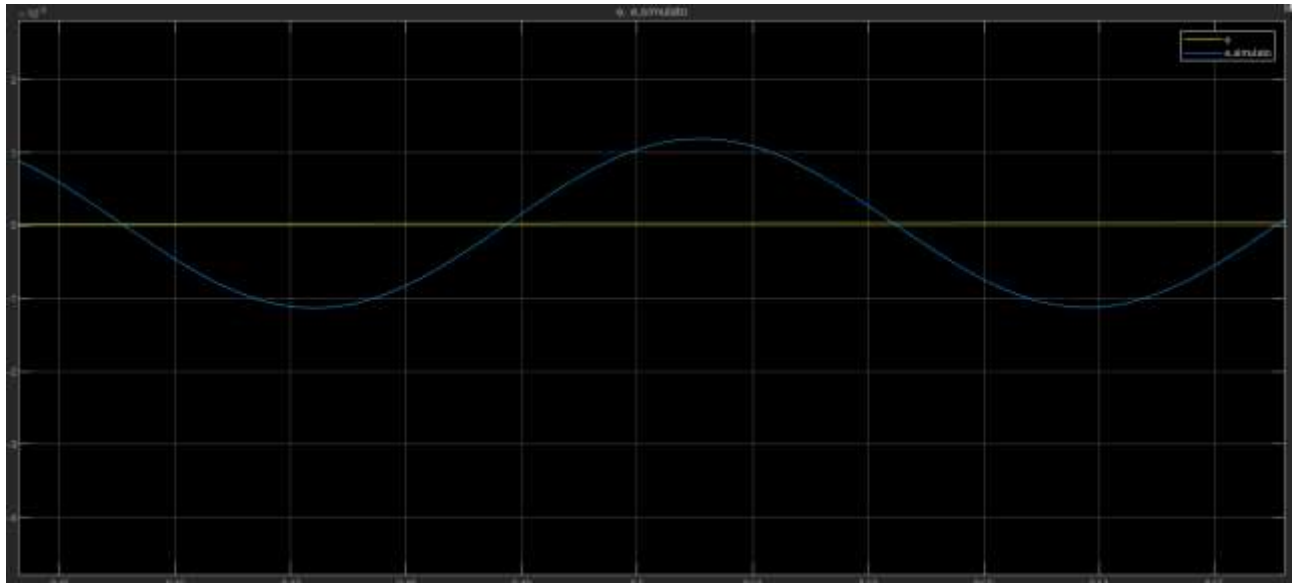
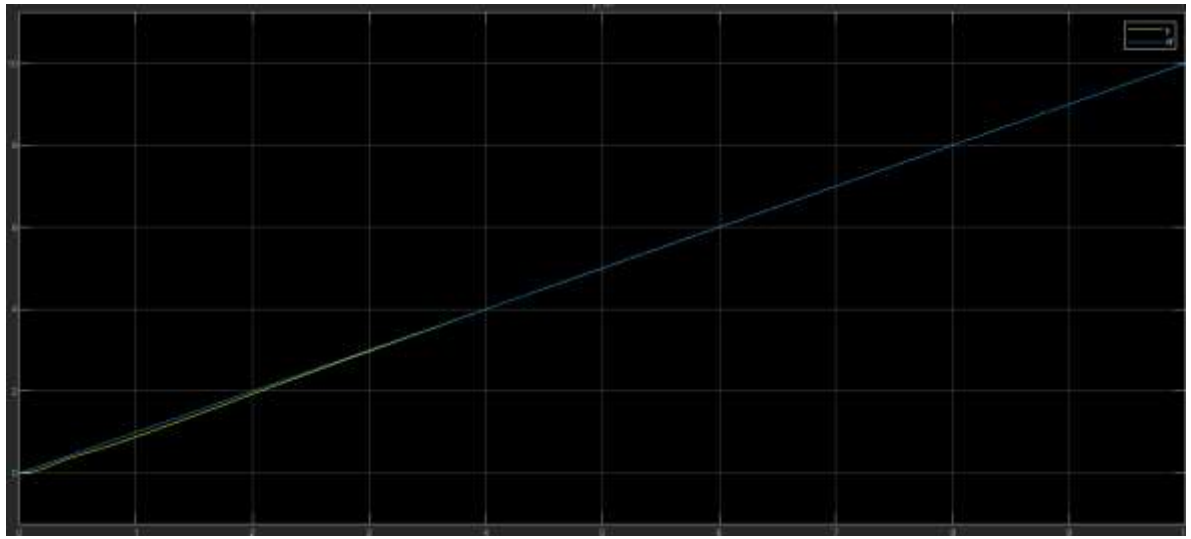


Figure 22 ↑ e 23 →



Precisione a regime



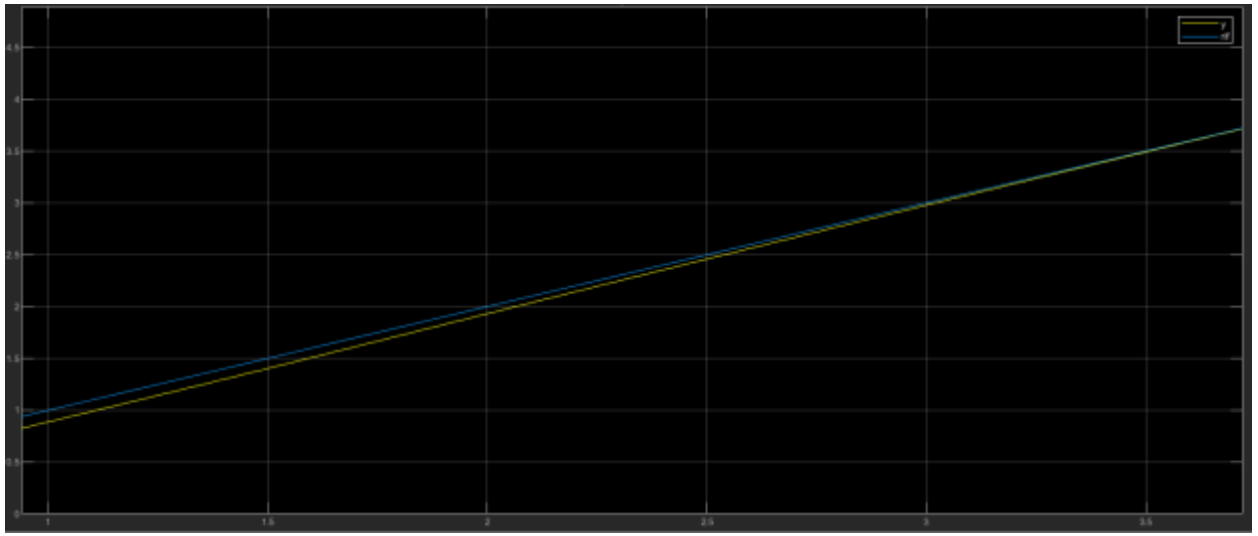


Figure 24 e 25

Effetti della saturazione

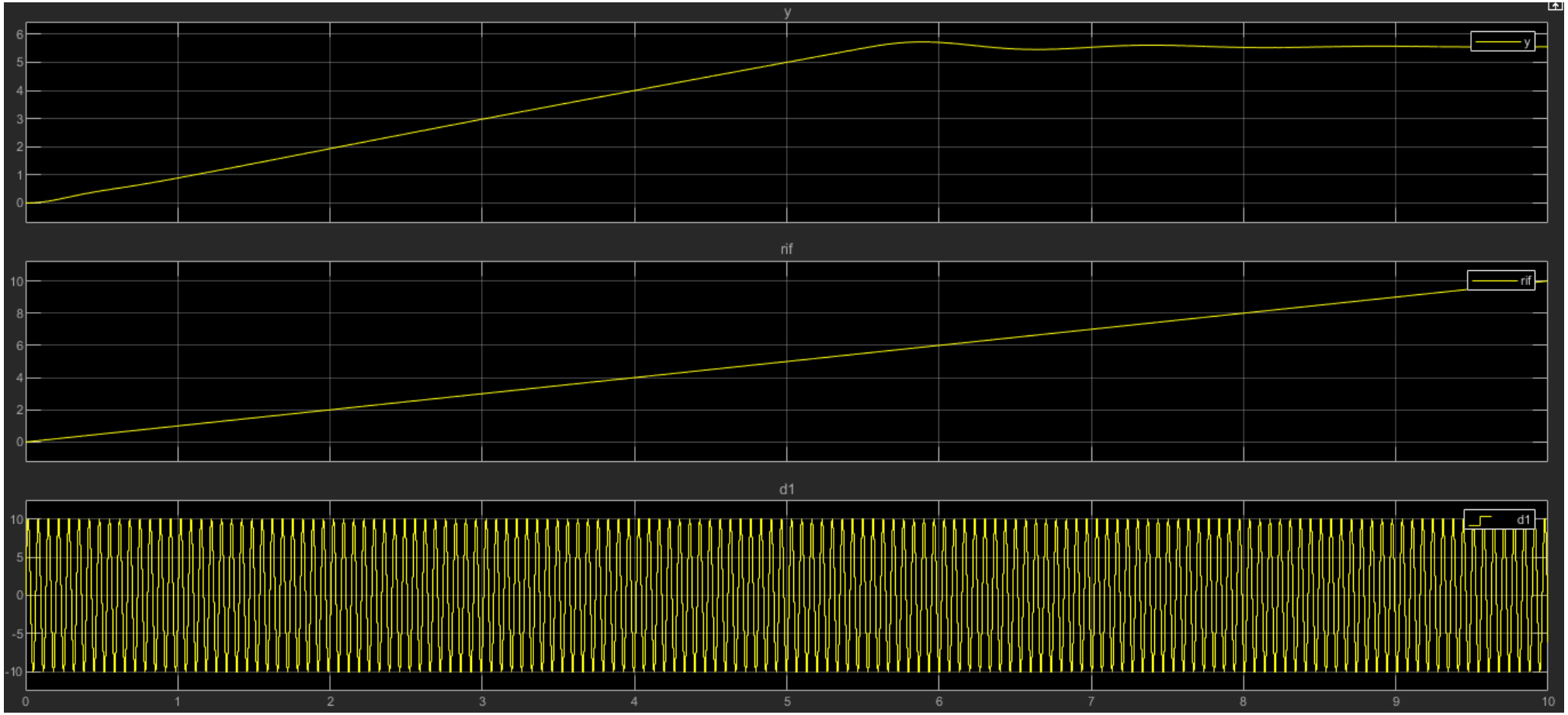


Figura 26

Effetti della saturazione

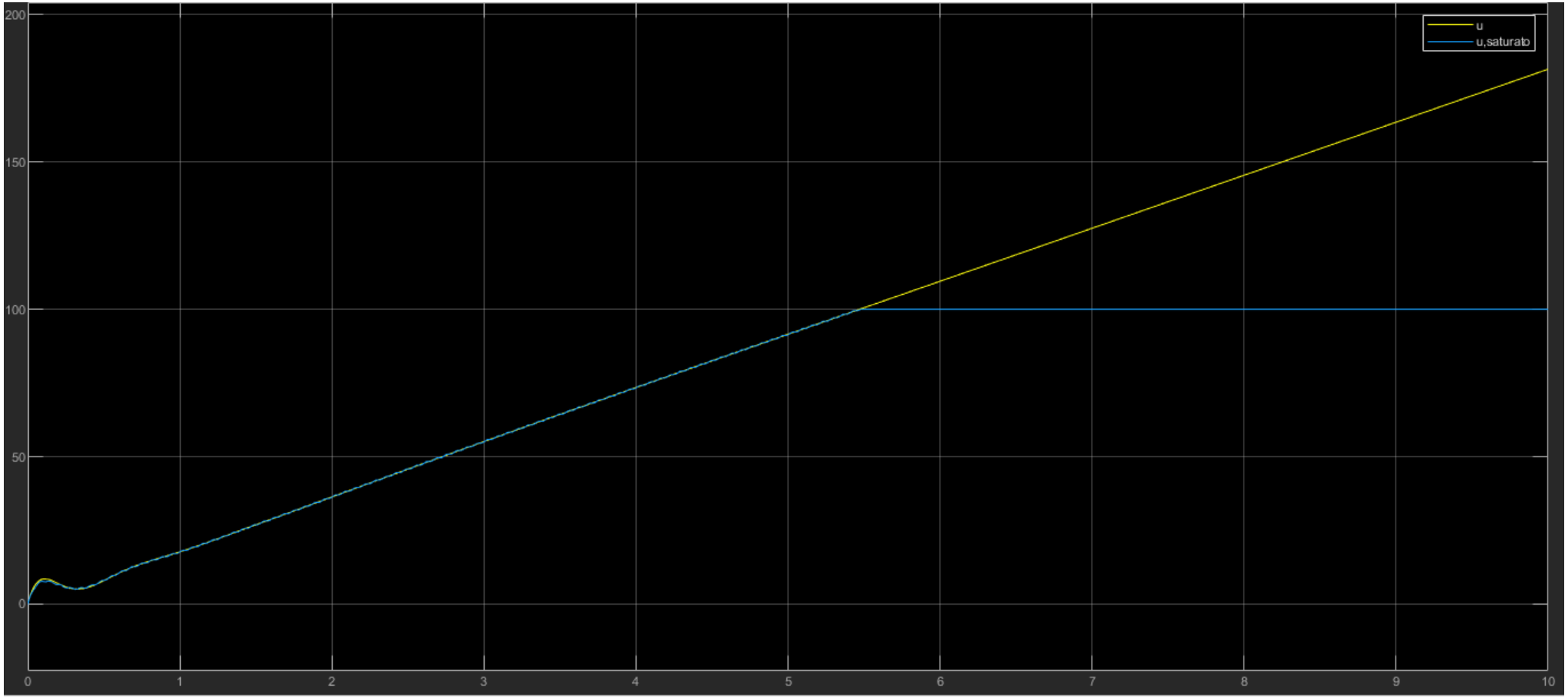


Figura 27

Effetti della saturazione

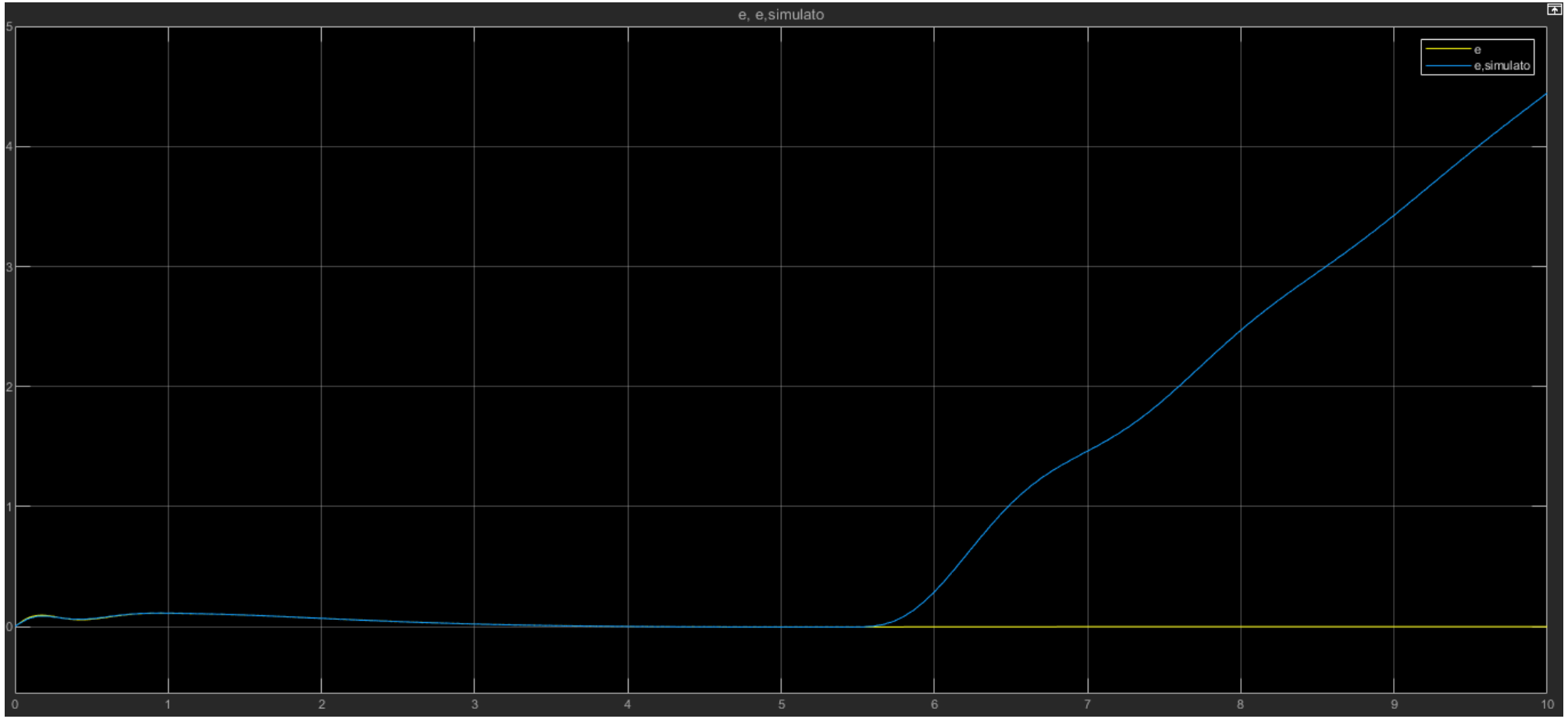


Figura 28

Effetti della saturazione

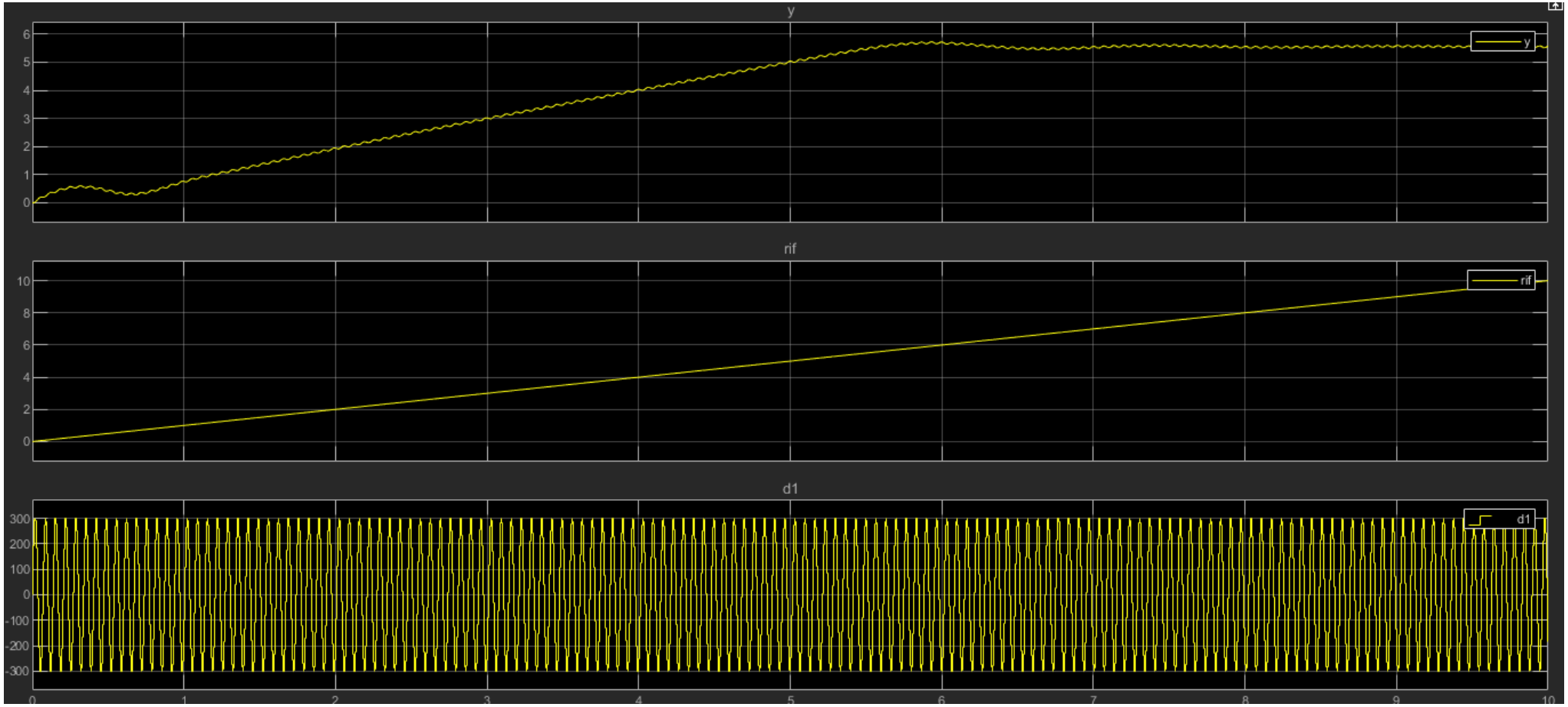


Figura 29

Effetti della saturazione

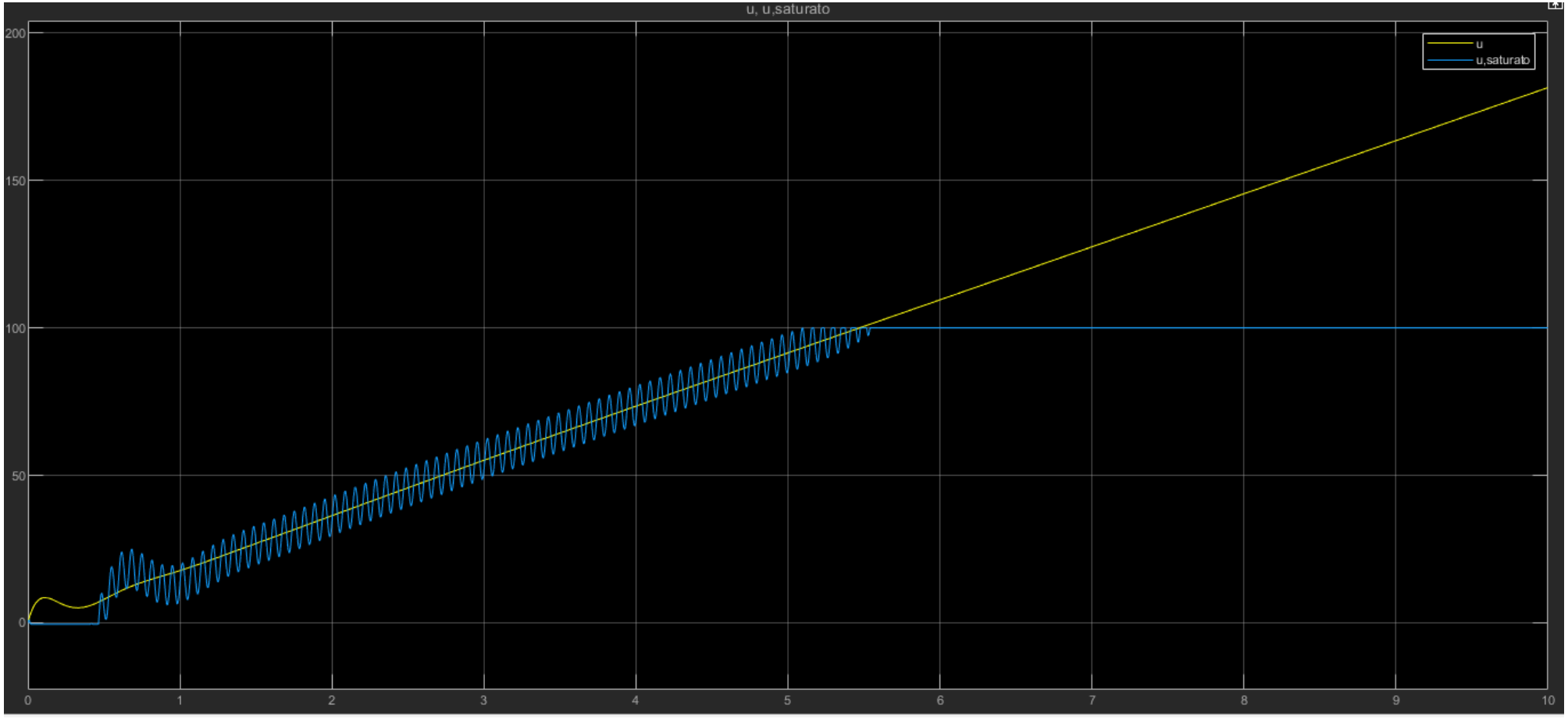


Figura 30

Effetti della saturazione

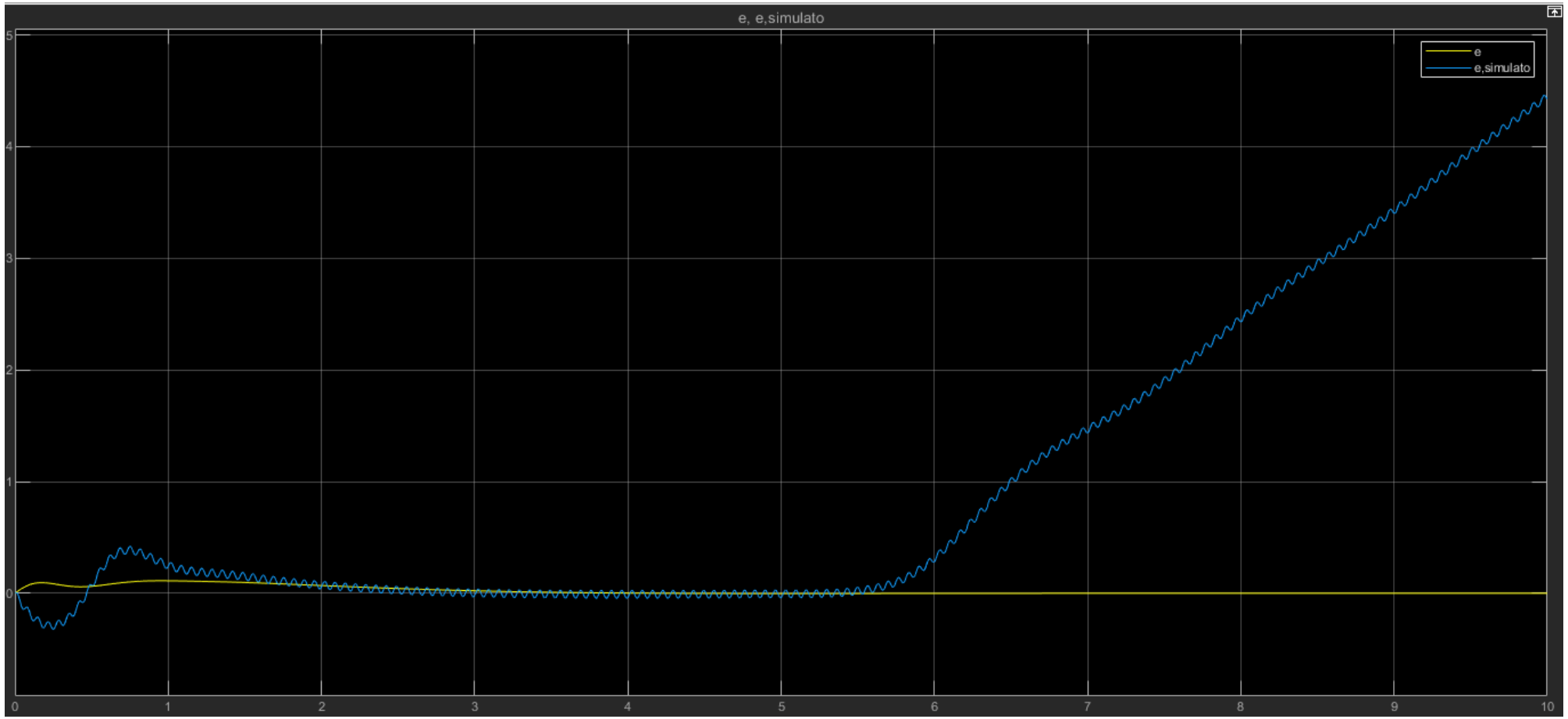


Figura 31

Fine
Presentazione